

Directed Research Project (DRP) Proposal Form

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Faculty and Program: Faculty of Architecture; Undergraduate Program in Architecture

Position: Visiting Faculty

Title of the DRP: Democratizing Residential Architecture in the digital age: Exploring Mass Customization as a Solution for Efficiency and Individualization

No of students the guide is accepting in this DRP: 2

Introduction about the proposed research

Architecture, traditionally considered a service industry, is witnessing a paradigm shift in response to the highly individualized society we live in today. Residential architecture, in particular, offers two contrasting typologies: the 'standardized' typology focused on mass production through simple repetition, and the 'tailor-made' typology emphasizing unique designs and customization.

However, as the demand for custom-built environments grows and project timelines shrink, both approaches face challenges in meeting the evolving needs of clients. While the former struggles with identity, the latter with efficiency. Concurrently, digital technologies offer opportunities to enhance architectural design and execution processes.

This research seeks to bridge the gap between the two contrasting typologies by offering a hybrid approach that combines the efficiency of mass production with the benefits of customization. By leveraging digital technologies and computational design methodologies, architects can optimize the design and construction processes, making them more adaptable, efficient, and responsive to the unique needs and preferences of clients. Through user-centric design strategies, this research aims to empower individuals to actively participate in shaping their living spaces and foster a sense of ownership. This research proposal aims to explore the concept of architecture as a product and leverage the methodologies of mass customization to address these challenges. By doing so, it aims to contribute to the evolution of the field, enhance efficiency and enable individualization.

Aim:

The aim of this research is to develop a choice-based computational framework to enhance efficiency and enable individualization in the design of residential buildings through interactive design tools.

Objectives:

1. Study existing G, G+1 and G+2 residential typologies and establish relationships (rules) between their constituent elements.
2. Develop a system for mass customization based on different spatial qualities and design approaches.
3. Develop an interactive algorithm capable of generating fabrication drawings based on user inputs.

Expected Outcomes:

This research is expected to yield a methodology and a codebase that applies mass customization ideologies to architecture, specifically focusing on residential buildings. The computational framework developed will provide architects and users with a tool to collaboratively generate and execute designs, leading to greater efficiency, individualization, and user satisfaction in the residential building sector.

By exploring the potential of architecture as a product and harnessing the power of computational design, this research aims to remodel the way residential buildings are conceived, designed, and constructed. The proposed computational framework will enable architects and non architects to tailor designs to meet the unique preferences and needs of clients while ensuring efficient execution within compressed timelines. Ultimately, this research seeks to enhance user agency, foster innovation in architectural practices, and contribute to the evolution of computational architectural design as a field.

Schedule of the DRP (weekly for 16 weeks)**Week 1: Literature Review**

Review relevant literature on mass customization in architecture
Identify key concepts, theories, and methodologies
Summarize and analyze existing research studies

Week 2: User Requirements Analysis

Define and categorize user requirements for residential buildings
Conduct interviews or surveys to gather user preferences and needs
Analyze the collected data and identify patterns and trends

Week 3: Space Planning Analysis

Study existing G+1 residential typologies
Identify and establish relationships (rules) between constituent elements
Analyze space planning strategies and their impact on customization options

Week 4: Rule-Based Construction System Design

Design a rule-based construction system that enables flexible customization options
Define parameters, constraints, and decision-making rules
Develop a conceptual framework for the rule-based system

Week 5: Algorithm Development - Part 1

Develop an interactive algorithm to generate space layouts based on user input choices
Implement algorithms for space optimization and customization
Test and refine the algorithm using sample scenarios

Week 6: Algorithm Development - Part 2

Continue refining and improving the interactive algorithm
Integrate feedback and make necessary adjustments
Conduct thorough testing and validation of the algorithm

Week 7: Workflow Design

Design an automated drawing board-to-consumer workflow
Define the stages and processes involved in the workflow

Identify potential bottlenecks and develop strategies to streamline the workflow

Week 8: Code Implementation

Translate the conceptual framework, algorithm, and workflow design into code

Develop software or computational tools for the choice-based computational framework

Debug and optimize the code for efficiency and usability

Week 9: Prototype Development

Build a working prototype of the choice-based computational framework

Test the prototype with sample residential building scenarios

Collect feedback from users and make necessary refinements

Week 10: Data Analysis and Evaluation

Analyze the results from testing and user feedback

Evaluate the effectiveness and efficiency of the choice-based computational framework

Identify strengths, weaknesses, and areas for improvement

Week 11: Methodology Documentation

Document the methodology and processes involved in the choice-based computational framework

Provide clear instructions and guidelines for users and practitioners

Include examples and case studies to illustrate the application of the framework

Week 12: Results Analysis and Discussion

Analyze and interpret the outcomes of the research

Discuss the implications and significance of the findings

Compare the results with existing literature and theories

Week 13: Conclusion and Recommendations

Summarize the main findings of the research

Draw conclusions based on the results

Provide recommendations for future research and practical applications

Week 14: Paper Writing - Introduction and Methodology

Begin writing the research paper

Draft the introduction section, including background and research objectives

Describe the methodology in detail, explaining the steps and processes involved

Week 15: Paper Writing - Results and Discussion

Write the results section, presenting the findings of the research

Discuss the results in light of the research objectives and existing literature

Analyze the limitations and propose future directions for research

Week 16: Paper Writing - Conclusion and Finalization

Write the conclusion section, summarizing the key findings and contributions

Proofread and edit the entire research paper

Finalize the paper, ensuring coherence and clarity in the content and structure

Relevant literature

Zoa Li, Brief analysis of spatial constitution and functional organization of museum; A case study on museums in Hefei City

Abdulmalik Abdulmawla, Decodings-paces Toolbox for Grasshopper

Hensel, M. (2013) Performance-Oriented Architecture: Rethinking Architectural Design and the Built Environment

Nourian, P., Rezvani, S., Sariyildiz, S. (2013). A Syntactic Design Methodology . Proceedings of 9th Space Syntax Symposium.

Rashid M, Kampschroer K, Wineman J, Zimring C (2006) Spatial layout and face-to-face interaction in offices - A study of the mechanisms of spatial effects on face-to-face interaction, Environment and Planning B Planning and Design · November 2006, DOI: 10.1068/b31123 · Source: RePEc

Spatial Accessibility & Human Behaviour - Designing places for People, Space Syntax - Christchurch (2013).
URL: [http:// www. spacesyntax. com/](http://www.spacesyntax.com/)

Michael J. Ostwald, Michael J. Dawes (2018) 'Isovist: Spatio-Visual Mathematics in Architecture.
URL: <https://link.springer.com/>

OECD (2016) Family Indicators (Edition 2016), OECD Social and Welfare Statistics (database), DOI: <http://dx.doi.org/10.1787/a6b32120-en> (Accessed on 20 April 2017)

Kieran S, Timberlake J (2004) Refabricating Architecture: How Manufacturing Methodologies are Poised to Transform Building Construction, McGraw Hill, NY

Pine B, (1993) Mass Customization: The New Frontier in Business Competition, Harvard Business School Press, Boston

Blecker T, Friedrich G (2006) Mass Customization: challenges and solutions, Springer, New York

Madrazo L, Sicilia A, González MG, Cojo AM (2009) Barcode housing system: Integrating floor plan layout generation processes within an open and collaborative system to design and build customized housing, CAAD Futures: 656- 670

Friedman A, Sprecher A, Mohamed BE (2013) A Computer-Based System for Mass Customization of Prefabricated Housing. Open house International 36(1): 20-30

Silveira GD, Borenstein D, Fogliatto FS (2001) Mass Customization: Literature review and research directions. International Journal of Production Economics: 1-13

Lo TT, Schnabel MA, Gao Y (2015) ModRule, A User-Centric Mass Housing Design Platform. CAAD Futures: 236-254

Lo TT, Schnabel MA, Moleta T (2017) Gamification for User-Oriented Housing Design: A Theoretical Review. CAADRIA: 63-73

Friedman A (2011) Decision Making For Flexibility in Housing, The Urban International Press, UK

Duarte JP (2001) Customizing Mass Housing: A Discursive Grammar for Siza's Malagueira Houses, MIT, Massachusetts

How is this DRP aligned with the long-term research direction of the guide?

Computational methodologies have been my focus of study, teaching and practice since the last 5 years and modular architecture for even longer. Space planning has interested me in particular since it is a fairly complex process and we often struggle to realize designs that can successfully convert qualitative aspects into tangible outcomes. I believe, data driven processes can generate unique design solutions that veer away from conventional thinking and offer newer ways of achieving the same design intent at a faster pace. In 2020, I assisted a studio at CEPT “DIGITAL PROCESSES AND TECHNIQUES IN DESIGN” which attempted the same for various institutional buildings like banks, libraries, theaters etc.

My own research on the topic started with my undergrad thesis in 2015 that focused on developing an adaptive architectural system. During my post graduation thesis, I developed a process by which spatial layouts of hypermarkets can be derived on the basis of boundary conditions provided. The intent was to use computational tools to improve the space quality and allow for innovative compositions as opposed to the current standard models. Currently, as I lead a StartUp about design innovations for the emerging trends of the building technology, our focus has been to productize architecture through building systems and automate the processes involved.

During Monsoon 2023, my Directed research Program aimed at developing an algorithm of educational institution typology. To further this research, I would like to conduct a study on developing an algorithm for spatial layouts of a residential typology using a modular system.

Student expenses:

None

External funding:

Is external funding available for the DRP? No

Only for funded DRPs (where external funding is available for the research)

Available stipend per student in case of Funded DRP: Rs. 40,000 per student for the full semester to be transferred after successful completion of the semester. In case the DRP is left midway, students will not be eligible to receive any stipend.

Details required:

Name of the organization:

Name of the contact person:

Contact details: email and telephone no.

Recommended by:

Approved by:

DRP Committee
(Faculty in-charge + Program Chair)

Dean, Faculty of Architecture